

# Intelligent Support Ticket Resolution Assistant

A telecom support agent pastes a customer complaint and gets labels, the most similar past cases, and a drafted step-by-step fix in which every step names the ticket or article it came from. It searches by meaning, not only by keyword, and says “escalate” when it finds nothing good enough.

Unnathi Vinod · [github.com/unnathivinod/ticket-resolution-assistant-](https://github.com/unnathivinod/ticket-resolution-assistant-)

## What the agent gets in one screen

1. **What it is:** category, product, severity and sentiment.
2. **What was done before:** similar past tickets and knowledge-base articles, found by meaning and by keyword.
3. **What to do now:** a drafted fix. Every step cites its source, and each citation is checked.

## The numbers, measured by scripts in the repository

### 55% vs 40%

the right ticket ranked first: search by meaning against keyword search

### 100%

of steps backed by their source, in answers to known problems

### 77%

of off-topic questions stopped before any language model is asked

### 15 of 20

known problems answered correctly by the hosted model on Groq (11 by the local one)

### 71%

of simulated outages noticed, with 2% false alarms

### 367 + 24

fast tests and tests against the running system, run on every push

## What the brief asked for, and where it is

| DELIVERABLE                      | WHERE                                                                                                        |
|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Architecture diagram             | Pages 3 and 4 here; <code>docs/ARCHITECTURE.md</code> , section 3                                            |
| Full executable code             | The repository. Five commands start everything (page 10)                                                     |
| Additional exploration           | Reply to the customer, incident detection, sign-in with roles and cases, new-class discovery (pages 5 and 6) |
| Evals on system health           | Six eval scripts, 19 tested alerts, a 28-panel dashboard (pages 7 and 9)                                     |
| Production scale considerations  | Page 9; <code>docs/ARCHITECTURE.md</code> , section 9                                                        |
| Evolving data and ticket classes | New data is searchable in seconds; new classes are proposed and approved (page 6)                            |

# What the agent sees

One complaint, pasted as the customer wrote it. Labels, a cited fix and the sources come back on one page.

Resolve a complaint

Paste what the customer wrote, then press Resolve

My broadband drops every evening around 8 and I've already restarted the router twice, I work from home and this is costing me

Resolve →

CATEGORY

Connectivity intermittent

92% of similar tickets agree

PRODUCT

Broadband

100% of similar tickets agree

SEVERITY

High

Business impact, money pressure

SENTIMENT

Neutral

No strong tone found

Suggested resolution

✓ 4 of 4 steps backed by a source

Give feedback

LIKELY CAUSE

The likely cause is peak-hour congestion on the local exchange causing the line to lose sync between 7pm and 10pm.

Already tried, not suggested again

Restarted the router twice

1

Run a remote line test and compare sync speed and error counts between daytime and 7pm to 10pm.

From KB-001 T-000481 T-000505 T-000511

2

Check the exchange utilisation report for the customer's cabinet to confirm peak-hour congestion.

From KB-001 T-000481 T-000505 T-000511

3

Change the router Wi-Fi channel to a less crowded one to rule out local interference.

From KB-001 T-000481 T-000505 T-000511

4

If the cabinet is over capacity, raise a capacity ticket with the network team and give the customer the expected upgrade date.

From KB-001 T-000481 T-000505 T-000511

Review the draft, then record how the case ended.

→ Escalate to second line

✓ Mark as resolved

Reply to the customer

A message you can edit and send, built only from the checked steps.

Draft reply to customer

Sources found

5 - best match first - click a row to read it

|          |             |                                                                              |      |          |
|----------|-------------|------------------------------------------------------------------------------|------|----------|
| KB-001   | Article     | Evening broadband drops caused by peak-hour congestion                       | 0.80 | Used     |
| T-000481 | Past ticket | Evening broadband drops caused by peak-hour congestion                       | 0.91 | Used     |
| T-000505 | Past ticket | Evening broadband drops caused by peak-hour congestion                       | 0.91 | Used     |
| T-000511 | Past ticket | Evening broadband drops caused by peak-hour congestion                       | 0.89 | Used     |
| KB-002   | Article     | Random broadband disconnects caused by faulty line filter or internal wiring | 0.76 | Not used |

The Resolve page. "Restarted the router twice" is recognised as already tried and is not suggested again.

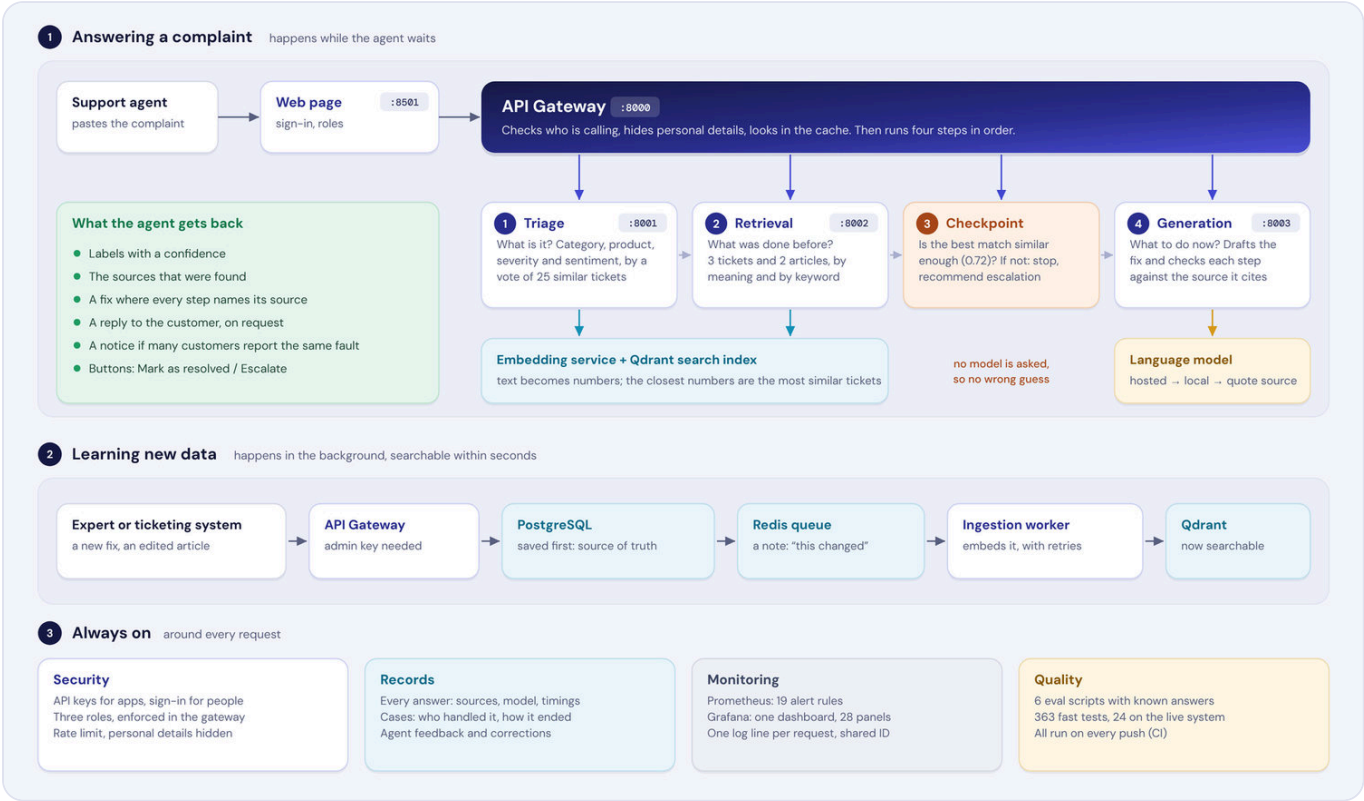
- **Labels** come from a vote among the most similar past tickets, each with a confidence.
- **Every step** shows the tickets and articles it was taken from. Steps that fail the source check are flagged.
- **The agent decides.** Mark as resolved or escalate; the assistant only suggests.

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# How it works

Two paths: answering a complaint while the agent waits, and learning new data in the background.



The path of a complaint, the path of new data, and what runs around every request.

| STEP | WHAT HAPPENS                                                                                 | TYPICAL TIME                             |
|------|----------------------------------------------------------------------------------------------|------------------------------------------|
| 1    | The gateway checks the caller and masks emails, phone and account numbers                    | instant                                  |
| 2    | Triage labels the complaint by a vote among the most similar past tickets                    | under 1 s                                |
| 3    | Retrieval finds the 3 closest tickets and 2 closest articles                                 | under 1 s                                |
| 4    | If nothing is similar enough, it stops here and recommends escalation                        |                                          |
| 5    | The language model drafts a fix. Every step must cite a source, and each citation is checked | 31 s local (Ollama), 1.4 s hosted (Groq) |

# Architecture

Seven small services behind one gateway, with ready-made infrastructure for search, storage, caching and monitoring.



Every request enters through the gateway, which calls the services in the order of the numbers. The dashed line is the queue for new data.

| SERVICE          | PORT | ITS ONE JOB                                                                                                 |
|------------------|------|-------------------------------------------------------------------------------------------------------------|
| Gateway          | 8000 | The only front door: API keys, sign-in and roles, rate limit, masking, cache, checkpoints, audit log, cases |
| Triage           | 8001 | Category, product, severity, sentiment. Says “unknown” when unsure                                          |
| Retrieval        | 8002 | Finds the most similar tickets and articles; counts similar recent complaints                               |
| Generation       | 8003 | Builds the prompt, calls the model, checks the answer. Backup model, then quotes the source                 |
| Embedding        | 8004 | Turns text into numbers that capture meaning, and into keywords                                             |
| Ingestion worker |      | Keeps the search index in step with the database, with retries and a safety sweep                           |
| Web page         | 8501 | Sign in, resolve, draft the reply, record the decision, list the cases, give feedback                       |

PostgreSQL is the source of truth, Qdrant is the search index (rebuildable at any time), Redis holds the cache, the rate-limit counters and the queue. The language model is a setting: a hosted model first, local Ollama as backup, and no paid API is needed to run the project.

# Beyond the brief

Four additions, each built on the parts the brief asked for.

## 1. A reply to the customer, in one click

Written only from the steps that passed the source check. It apologises if the customer was upset, does not ask them to repeat what they already tried, and contains no ticket numbers.

**Reply to the customer** ✓ Built from 4 checked steps Edit

Hello,

We're looking into the evening drop-outs you're experiencing. First, we'll run a remote line test to compare your sync speed and error counts during the day and between 7 pm and 10 pm. Next, we'll check the exchange utilisation report for your cabinet to see if peak-hour congestion is the cause. We'll also change your router's Wi-Fi channel to a less crowded one to rule out local interference. Finally, we may move you to a less loaded port or apply a congestion-relief profile.

We'll keep you updated as we progress. Thank you for your patience.

Kind regards,  
[Agent name]

Read it before sending. The copy button is at the top right of the message.

[Write it again](#)

The drafted reply, ready to edit and copy.

## 2. Possible service incident

One customer is a ticket; many customers with the same fault within minutes is an outage. Complaints that mean the same and arrive close together raise a notice on the page and an alert for the person on call.

**Resolve a complaint** Paste what the customer wrote, then press Resolve

The internet has been down since this morning in Adyar and my router is showing a red light.

**Resolve** →

**Possible service incident.** 7 similar complaints were received in the last 30 minutes. Hide

25 min ago Internet is not working in Adyar since early this morning, the router has a red light.

25 min ago Since this morning the internet is completely down in Adyar and the router light has turned red.

25 min ago My broadband has been down since this morning in Adyar, the light on the router is red.

**CATEGORY**  
**Connectivity outage**  
85% of similar tickets agree

**PRODUCT**  
**Broadband**  
74% of similar tickets agree

**SEVERITY**  
**High**  
No urgency signals

**SENTIMENT**  
**Neutral**  
No strong tone found

The notice, opened to show the other complaints.

### 3. Sign-in, roles and cases

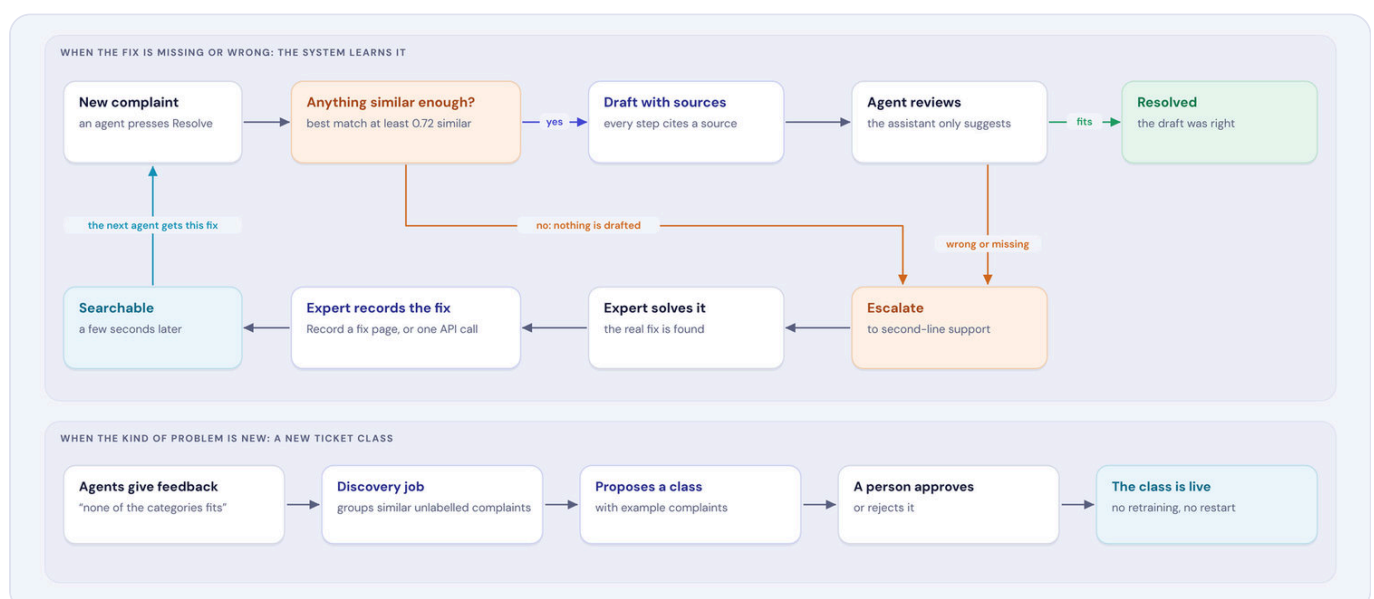
People sign in. An agent sees only the cases they handled; a second-line expert or an engineer sees the whole desk. The rule is enforced in the gateway, not in the page. Every case records what the assistant suggested and what the person decided, so the system measures how often its suggestions are followed.

| Cases                                                       |                                                                                  |                                         |                                     |                                                         |                     |            |
|-------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------|---------------------------------------------------------|---------------------|------------|
| Every complaint handled on the desk, and how each one ended |                                                                                  |                                         |                                     |                                                         |                     |            |
| HANDLED ON THE DESK<br><b>8</b><br>last 24 hours            | RESOLVED<br><b>6</b><br>75% of handled                                           | ESCALATED<br><b>1</b><br>12% of handled | OPEN<br><b>1</b><br>no decision yet | SUGGESTION FOLLOWED<br><b>86%</b><br>of 7 decided cases |                     |            |
| All                                                         | Open                                                                             | Resolved                                | Escalated                           | Last 24 hours                                           |                     |            |
| WHEN                                                        | COMPLAINT                                                                        | CATEGORY                                | SEVERITY                            | ASSISTANT SUGGESTED                                     | DECISION            | HANDLED BY |
| 1 min ago                                                   | This is getting ridiculous. My broadband keeps disconnecting every evening...    | Connectivity intermi...                 | High                                | ✓ Fix drafted                                           | ✓ Resolved          | Arun K     |
| 5 min ago                                                   | I got my new SIM yesterday and it still hasn't activated. I've followed every... | Activation provision...                 | High                                | ✓ Fix drafted                                           | ✓ Resolved          | Arun K     |
| 6 min ago                                                   | My new SIM has not activated since yesterday                                     | Activation provision...                 | Medium                              | ✓ Fix drafted                                           | Open                | Arun K     |
| 7 min ago                                                   | I was charged twice this month                                                   | Billing dispute                         | Medium                              | ✓ Fix drafted                                           | ✓ Resolved          | Arun K     |
| 40 min ago                                                  | My broadband drops every evening around 8 and I've already restarted the...      | Connectivity intermi...                 | High                                | ✓ Fix drafted                                           | ✓ Resolved          | Arun K     |
| 42 min ago                                                  | My Wi-Fi works perfectly when I wake up, but once everyone in my area gets...    | Slow speed                              | Medium                              | ✓ Fix drafted                                           | ✓ Resolved          | Priya S    |
| 44 min ago                                                  | wifi drop in adyar                                                               | Unknown                                 | Medium                              | ✓ Fix drafted                                           | → Escalated differs | Arun K     |
| 45 min ago                                                  | My broadband drops every evening around 8 and I've already restarted the...      | Connectivity intermi...                 | High                                | ✓ Fix drafted                                           | ✓ Resolved          | Priya S    |
| 8 cases. Personal details are masked.                       |                                                                                  |                                         |                                     |                                                         |                     |            |

The Cases page of an expert. One case is marked “differs”: a fix was drafted, and the agent escalated instead.

### 4. Learning what it did not know

When nothing fits, the complaint is escalated. The expert records the real fix on the page or by one API call, and it is searchable a few seconds later. When agents keep saying “none of the categories fits”, a job groups those complaints and proposes a new ticket class for a person to approve.



The loop for a missing fix, and the path to a new ticket class.

# What was measured

Every number comes from a script in the repository and can be reproduced. The test complaints use wording that never appears in the indexed tickets, so this is the hard case.

| QUESTION                                    | RESULT                                                                                                                                                                |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does search by meaning beat keyword search? | The right ticket comes first for <b>55%</b> of complaints, against 40% for keywords                                                                                   |
| How good are the labels?                    | Category 71%, product 78%, sentiment 83%, severity 59% exact and 93% within one level                                                                                 |
| How fast does new data arrive?              | A new ticket is searchable <b>a few seconds</b> after one API call. Nothing is retrained or restarted                                                                 |
| Can it learn a new kind of problem?         | Search: yes, from a handful of tickets. Category: 18 of 20 for one new class, 1 of 20 for one that overlaps existing classes                                          |
| Is every step backed by its source?         | <b>100%</b> of steps in answers to known problems                                                                                                                     |
| Are off-topic questions stopped?            | 77% by the similarity cut-off, before any model is asked                                                                                                              |
| Is an outage noticed?                       | Seven customers reporting one fault among twenty other complaints are flagged in <b>71%</b> of cases; 2% of quiet half hours are flagged by mistake                   |
| How much load does it carry?                | On one laptop, labels and sources for <b>8 to 9 complaints a second</b> with no errors, up to 20 agents at once. Typical response 0.2 s for one agent, 2 s for twenty |

## Two language models on the same 56 complaints

|                                                 | LOCAL LLAMA3.2:3B ON OLLAMA | HOSTED GPT-OSS-20B ON GROQ |
|-------------------------------------------------|-----------------------------|----------------------------|
| Known problems answered correctly (of 20)       | 11                          | <b>15</b>                  |
| When the search had found the right source (15) | 11 right, 3 mixed, 1 wrong  | <b>15 of 15 right</b>      |
| When the search had missed it (5)               | 5 wrong answers             | 3 wrong, 2 refused         |
| Problems with no fix in the knowledge base (6)  | 6 wrong answers             | 3 wrong, <b>3 refused</b>  |
| Off-topic questions that reached the model (7)  | 7 answered                  | <b>7 refused</b>           |
| Typical time per drafted answer                 | 31 s                        | <b>1.4 s</b>               |

With the larger model, every remaining wrong answer on a known problem is a search miss. The model is no longer the weak link; the search is. These are single runs on small groups: a direction, not exact rates.

# Design decisions

The main choices, what was considered instead, and why. The full list with the measurements is in `docs/DESIGN_DECISIONS.md`.

| DECISION          | CHOSEN                                                           | INSTEAD OF                                | WHY                                                                                                   |
|-------------------|------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Knowledge         | Search first, then let the model write only from what was found  | Fine-tuning a model on the tickets        | New tickets arrive every day. Search picks them up in seconds; retraining would not                   |
| Front door        | One gateway calls the services in order                          | The web page calls each service           | One place for security, limits, caching and logging                                                   |
| Labels            | A vote among the most similar past tickets                       | Asking the language model to classify     | Fast, explains itself with a confidence, and needs no retraining when a class is added                |
| When nothing fits | A similarity cut-off stops before the model                      | Always draft an answer                    | A confident wrong fix costs more than no fix                                                          |
| Storage           | PostgreSQL holds the truth, Qdrant is a search index             | Keeping everything in the vector database | The index can be rebuilt at any time, and a missed update is repaired from the database               |
| Cache             | Exact match on the masked complaint, one hour                    | A cache for similar complaints            | The same text always got the same sources. A similar-text cache can serve the wrong customer's answer |
| Language model    | Hosted model first, local model as backup, then quote the source | One model                                 | Fast and better when a key is set, yet the project runs with no key and no cost                       |

## Where the measurements changed the design

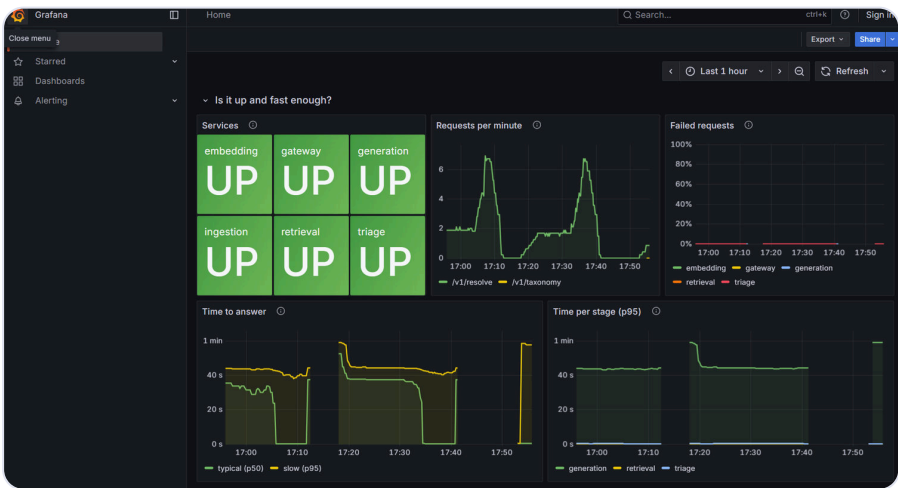
| PLAN                                              | WHAT THE MEASUREMENT SAID                                                               | WHAT CHANGED                            |
|---------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------|
| Hybrid search plus a reranker                     | Equal-weight hybrid was worse than meaning alone; the reranker added 900 ms for no gain | Meaning weighted 3x, reranker off       |
| Incident settings by a first guess                | 38% of outages noticed, 16% false alarms                                                | Settings picked by the eval: 71% and 2% |
| A prompt rule: escalate if the sources do not fit | Ignored 13 times out of 13                                                              | A similarity cut-off before the model   |

# Built for production

| CONCERN               | WHAT IS BUILT                                                                                                                                                                                             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A part fails          | Each service can fail without taking the rest down. Model down: the backup model answers, then steps are quoted from the source. Triage down: answer without labels. Cache or database down: still answer |
| Security and privacy  | API keys with separate agent and admin rights, a rate limit, sign-in with three roles, and personal details masked before anything is stored or sent to a model                                           |
| Data that changes     | New tickets, edited articles and new classes go live through the API, by a queue with retries and a safety sweep. Old cached answers are dropped automatically                                            |
| Trust in the answer   | Citations are checked against the sources, unsupported steps are flagged, and the customer reply uses only steps that passed                                                                              |
| Knowing it is healthy | 19 alert rules with their own tests, a 28-panel dashboard, a one-command health check, one log line per request with the same ID in every service                                                         |
| Knowing it is correct | Six eval scripts, 367 fast tests, 24 tests against the running system, CI on every push. In live use: how often agents do what the assistant suggested                                                    |

## At a larger scale

| AREA           | NOW (ONE LAPTOP)                                              | AT SCALE                                                                                                     |
|----------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Scaling        | One container per service: 8 to 9 requests a second, measured | Services hold no state: several copies behind a load balancer, starting with triage, retrieval and embedding |
| Language model | Small local model, or a hosted one                            | GPU serving or a hosted model; stream the answer                                                             |
| Search index   | One Qdrant node                                               | One million tickets is about 1.5 GB: still one node. Beyond that, sharding and replication                   |
| Ingestion      | Redis Streams, one worker                                     | More workers in the same group; Kafka at very high volume                                                    |
| Security       | Own sign-in, API keys in settings                             | Single sign-on, a secret manager, encryption at rest                                                         |



The monitoring dashboard: up and fast enough, answer quality, drift, language model and cache, new data.

# Known limits

- **Search is the weak link.** The right source is among those given to the model for about four complaints in five. Almost every wrong answer starts there.
- **It does not always know when it has no fix.** For a telecom problem the knowledge base does not cover, the local model drafts a confident wrong answer; the larger model refuses half of them. A person must review every draft.
- **The customer reply is not measured yet.** Rules check its form and it only receives checked steps, but its wording has no eval.
- **Incident detection only reads the complaint text.** It is a hint for a person, not an outage system.
- **Sign-in is basic.** Demo accounts with one public password, no single sign-on, no password reset.
- **The data is synthetic,** built from 40 hand-written problem scenarios. Real tickets are messier.
- **Small samples** in the answer eval (20 known complaints). The numbers show direction, not precision.

# Run it

You need Docker Desktop and Ollama. No paid API and no API key.

```
ollama pull llama3.2:3b
cp .env.example .env                                # Windows: copy .env.example .env
docker compose up -d --build
docker compose ps                                    # every service should show "healthy"
docker compose run --rm tools python scripts/seed.py
```

Then open `http://localhost:8501` and sign in. Three demo accounts share the password `demo1234` :

| USERNAME | ROLE               | SEES                                                |
|----------|--------------------|-----------------------------------------------------|
| priya    | Agent              | Resolve, Feedback, and only their own cases         |
| arun     | Second-line expert | The same, plus Record a fix and every agent's cases |
| meera    | Engineer           | Everything, plus the Monitoring link                |

# Where to read more

| DOCUMENT                 | WHAT IS IN IT                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------|
| README.md                | What it is, how to run it, what to try, what was measured                                  |
| docs/ARCHITECTURE.md     | The design in plain words, the diagrams, how each requirement is met, scale considerations |
| docs/DESIGN_DECISIONS.md | Each choice, the alternative, and the measurement behind it                                |
| docs/GUIDE.md            | Every address and command: API calls, evals, monitoring, tests                             |